

Product Requirements Document (PRD)

Project: Gradebook Revision

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1. Overview

Teachers reported several usability issues and missing features in the current LMS gradebook, particularly around assignment categorization, grade weighting, pacing guide integration, and grading workflows. This project defines a revised gradebook experience that allows teachers to customize grading categories, automate pacing-based scoring, and improve navigation and efficiency.

The revised gradebook will:

- Allow teachers to define assignment categories and weights per course section.
- Reflect those weights in automatic grade calculations.
- Improve visibility into student submissions and grading status.
- Support resubmission indicators and better filtering by section or course.

2. Problem Statement

The existing gradebook lacks flexibility for teachers to:

- Categorize assignments by type (e.g., Lab, Speaking Exercise, Attendance).
- Define their own grade weighting system per section.
- Filter, search, or navigate efficiently across students and assignments.
- Automatically calculate pacing-related zeros or manage resubmissions effectively.

These limitations lead to teacher frustration, inconsistent grading practices, and reduced accuracy in reporting student progress.

3. Goals and Success Metrics

Goals: - Improve grading flexibility - Increase efficiency in grading - Improve accuracy - Enhance user satisfaction

Success Metrics: - 90% of teachers successfully create custom grade weight configurations - Average grading time per assignment decreases by 25% - Grades correctly reflect pacing and category weights - Post-launch teacher satisfaction rating $\geq 4.5/5$

4. Stakeholders

Primary Users: Teachers managing courses and sections
Secondary Users: Academic administrators
Internal Stakeholders: Engineering, QA, Product

5. User Stories

Teacher: Gradebook Setup As a teacher, I want to assign weight percentages to different grading categories so that my course grades reflect my priorities. *Acceptance Criteria:* - Teacher can enter percentages across categories (quizzes, exams, labs, etc.) - Percentages must total 100%. - If weights are not defined, grades default to total points earned / total points possible.

Teacher: Assignment Setup As a teacher, I want to label assignments by category so the system can automatically apply correct weights. *Acceptance Criteria:* - Assignment creation includes a drop-down category selector. - Default category = "Assignment." - If no category is selected, it remains "Assignment."

Teacher: Filtering and Navigation As a teacher, I want to filter students by course or section and navigate quickly between submissions so I can grade efficiently. *Acceptance Criteria:* - Teachers can filter by Course > Section or select "All Sections." - Teachers can use arrow keys to move between students or assignments. - Teachers can search by student name or assignment title.

Teacher: Resubmissions and Grading Status As a teacher, I want clear visual indicators when students resubmit assignments so I don't miss new work. *Acceptance Criteria:* - Cells for new submissions display red until graded. - Resubmissions display yellow. - Graded items return to normal background color.

Teacher: Pacing Guide Integration As a teacher, I want missed pacing guide deadlines to automatically record zeros so that grades accurately reflect student progress. *Acceptance Criteria:* - If a pacing guide due date passes without submission, the assignment automatically receives a zero. - Zeros are included in total grade calculation until the assignment is submitted and graded.

6. High-Level Functional Requirements

- Category Setup: Add drop-down options for assignment types (Lab, Speaking & Listening Exercise, Fitness Log, TDI Attendance, Test Prep Attendance, Extended Assignment). - Weight Setup: Teachers can define category weights at the section level. Total must equal 100%. - Grade Calculation: Use teacher-defined weights; fallback to total points if no weights exist. - Filtering: Add optional section filter and course-wide view. - Search: Add search bar to find students or assignments. - Navigation: Support keyboard navigation and auto-fill functions. - Visual Indicators: Add resubmission and ungraded color states. - Pacing Integration: Auto-zero overdue assignments per pacing guide.

7. Dependencies & Constraints

- Dependencies: Existing assignment creation workflow; LMS database for submissions and pacing guides. - Constraints: Maintain backward compatibility with current grading schema; do not alter historical grade data.

8. Risks & Mitigation

- Teachers forget to assign weights → Default to points-based calculation. - Pacing guide data incomplete → Display warning for missing due dates. - Excessive complexity in setup → Use guided setup wizard and tooltips.

9. Future Enhancements

- Ability to import/export grading configurations across sections. - Analytics dashboard for pacing compliance. - Bulk-edit features for categories and weights.